Funded by
DFG Deutsche Forschungsgemeinschaft
German Research Foundation

Universität Potsdam

# Combining Satellite and In-Situ Data for Ecosystem Monitoring

February 22, 2026

# Welcome!

Dr Taylor Smith

Dr Bodo Bookhagen

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# Today's Schedule

**Day 1: Designing, building, and analysing in-situ sensor data in the Trishuli**

**09:30 – 10:00 Introduction from Kathmandu University**

**10:00 – 10:15 Introduction from Workshop Leaders**

**10:15 – 10:30 Collecting GNSS data in the Trishuli**

**10:30 – 10:45 Tea/Coffee Break**

**10:45 – 11:15 Processing GNSS data to estimate water vapor changes**

**11:15 – 12:30 Practical Example – Working with GNSS RINEX data**

**12:30 – 13:30 Lunch Break**

**13:30 – 14:00 Low-cost environmental stations in the Trishuli**

**14:00 – 14:30 Introduction to using camera data for pixel tracking and river-velocity estimation**

**14:30 – 14:45 Tea/Coffee Break**

**14:45 – 15:45 Practical Example – Measuring river flow velocity with a camera**

**15:45 – 16:00 Wrap Up and Further Resources**

# Long-term Snow Volume Trends

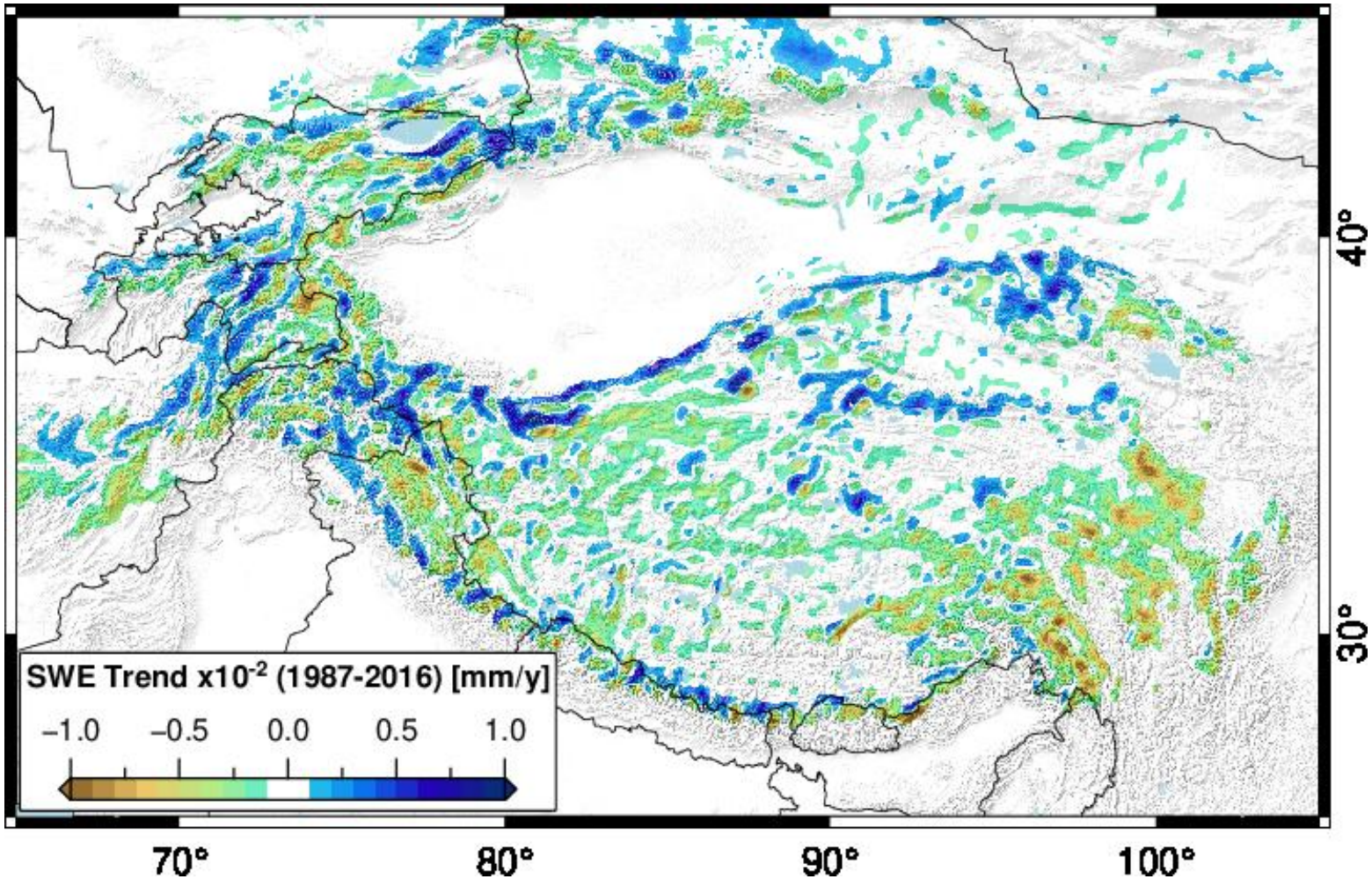
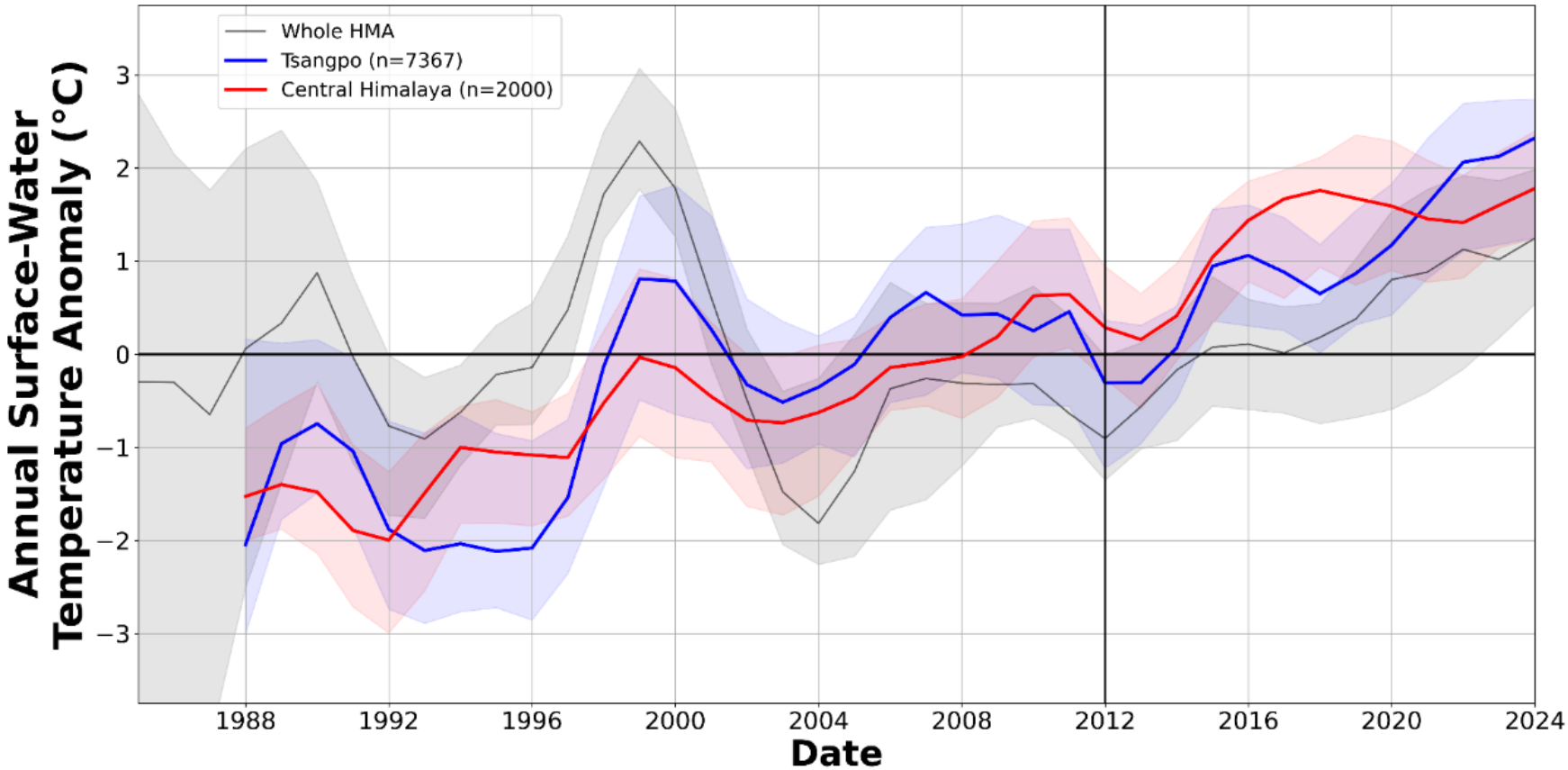



Image: Smith and Bookhagen, 2020

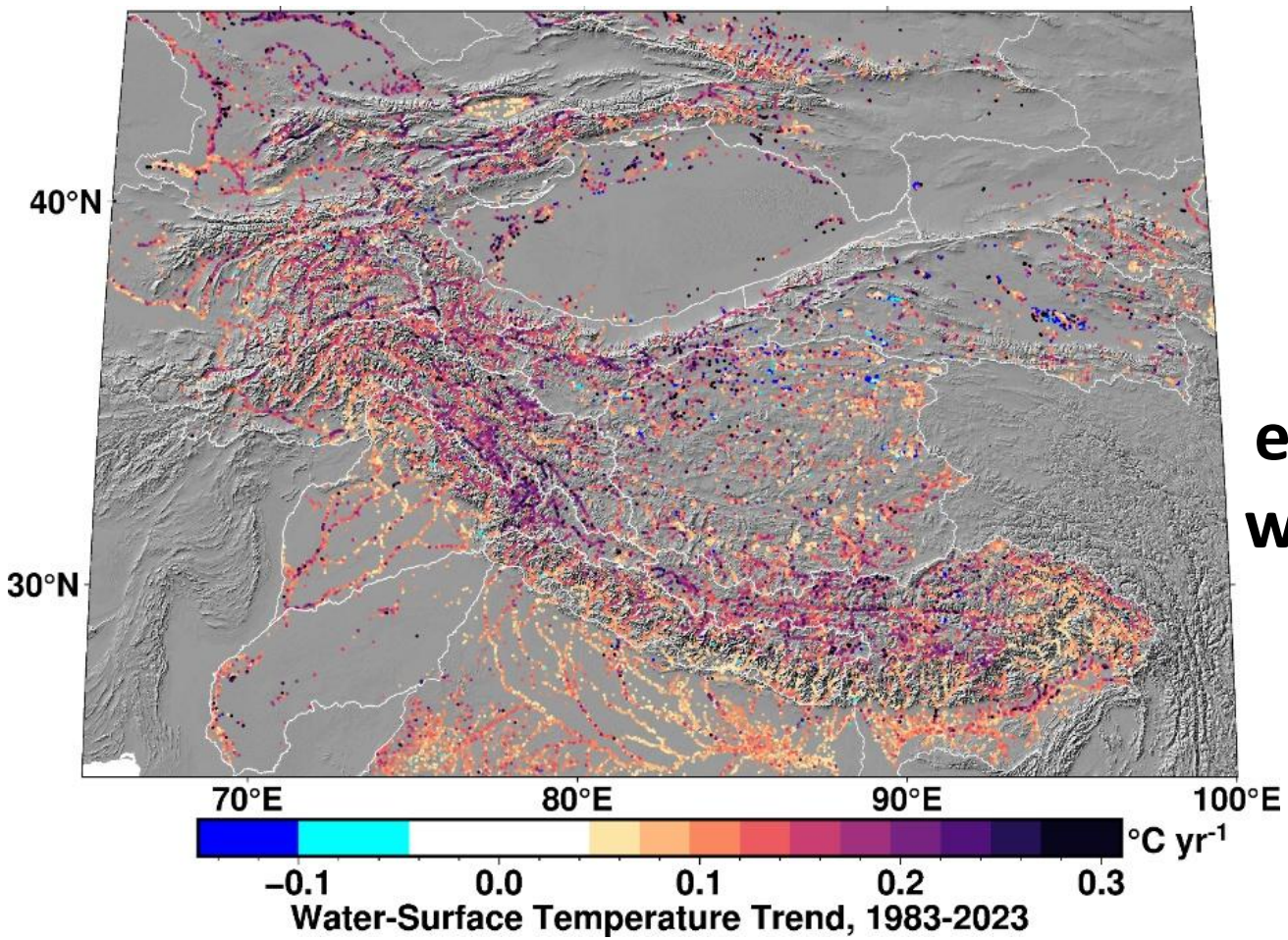
# Rising Water Temperatures




**River temperatures are rising steeply in many catchments**

Smith and Bookhagen, 2025

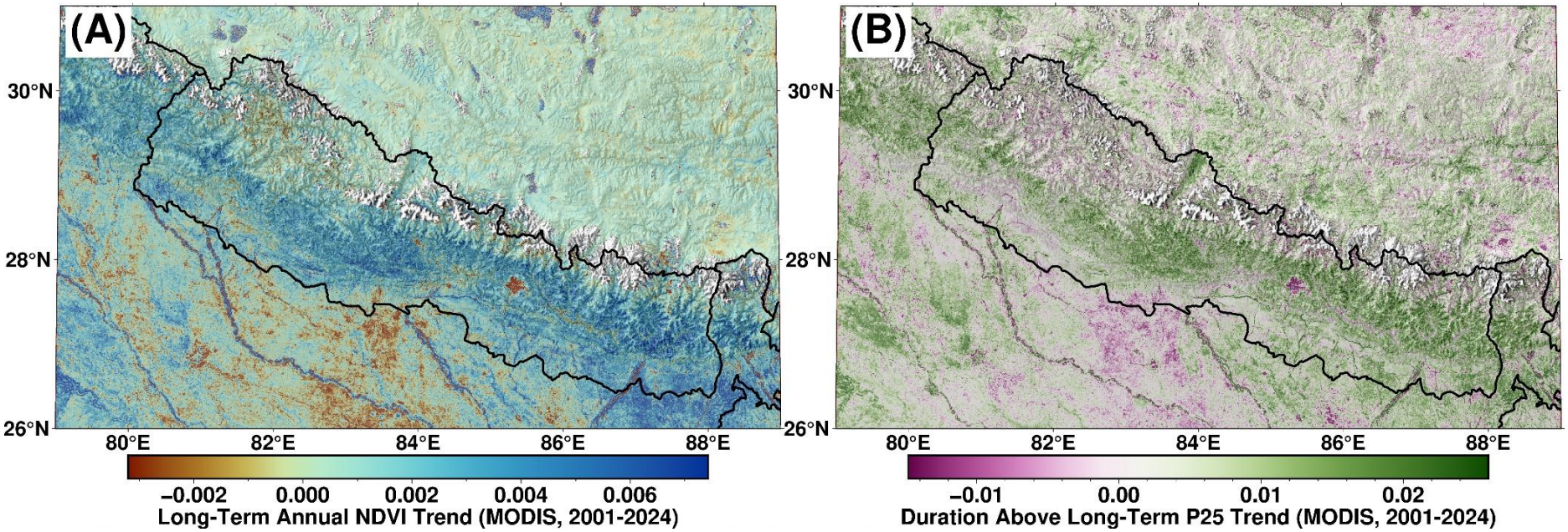
# Rising Water Temperatures




**The highest elevation rivers are warming the fastest**

Smith and Bookhagen, 2025

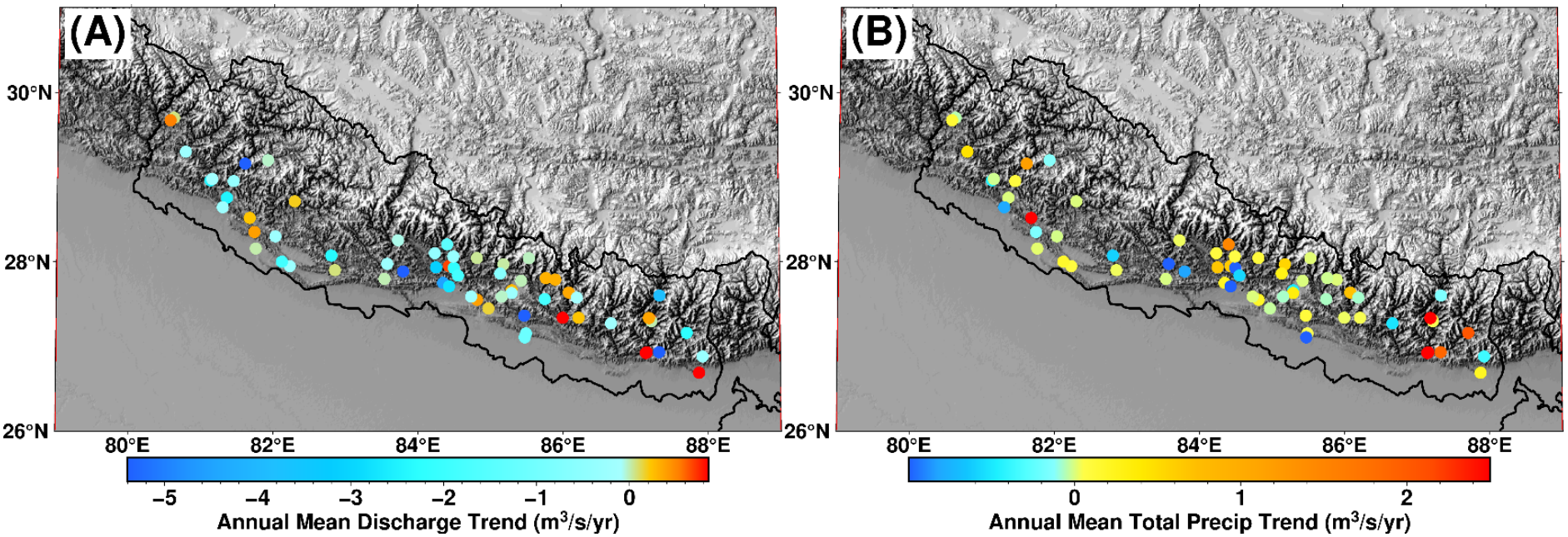
# Vegetation Increases




**Vegetation has gotten significantly greener – this is impacting water and sediment budgets**

Smith and Bookhagen, in prep

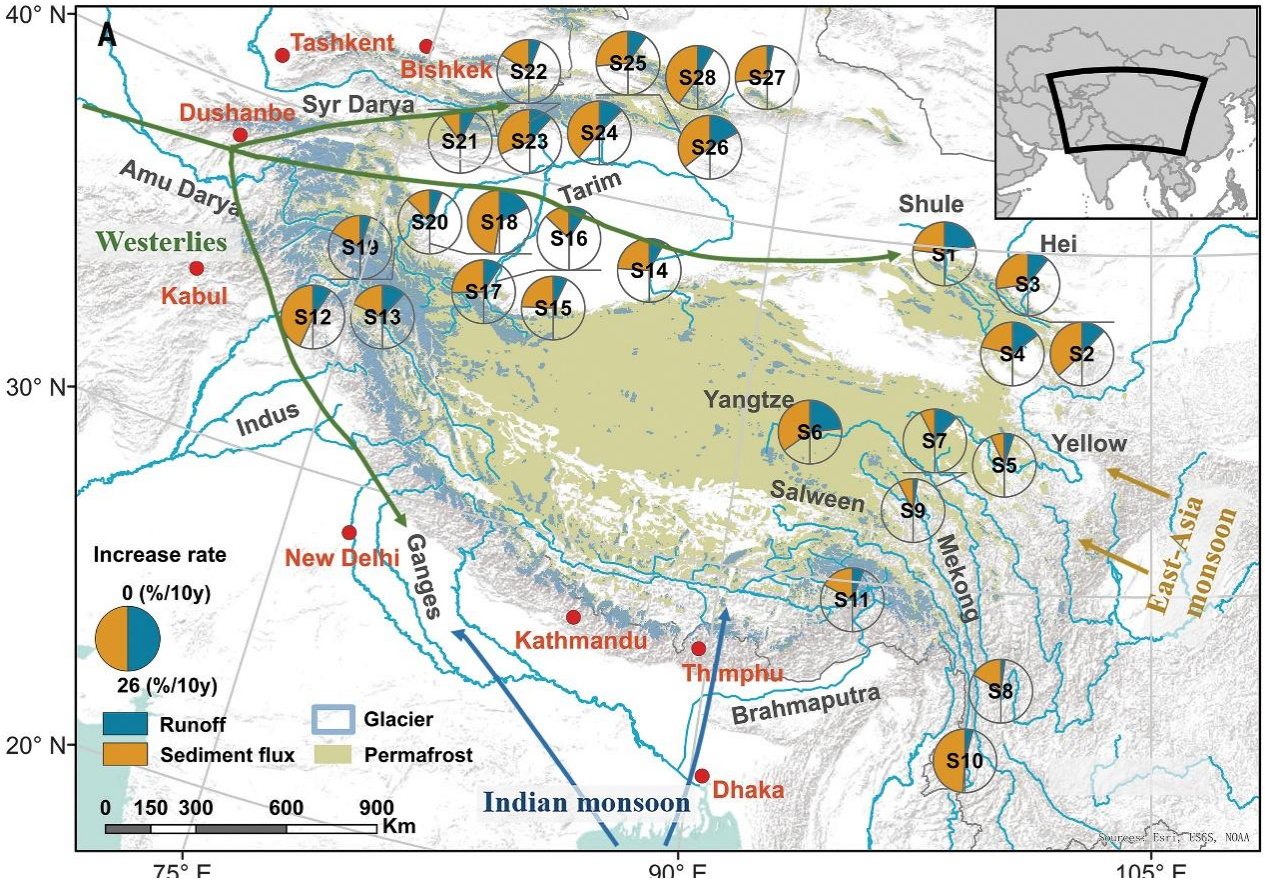
# Changing Discharge




**Discharge and precipitation trends are diverging – the relationship between rainfall and discharge is changing in many areas**

Smith and Bookhagen, in prep

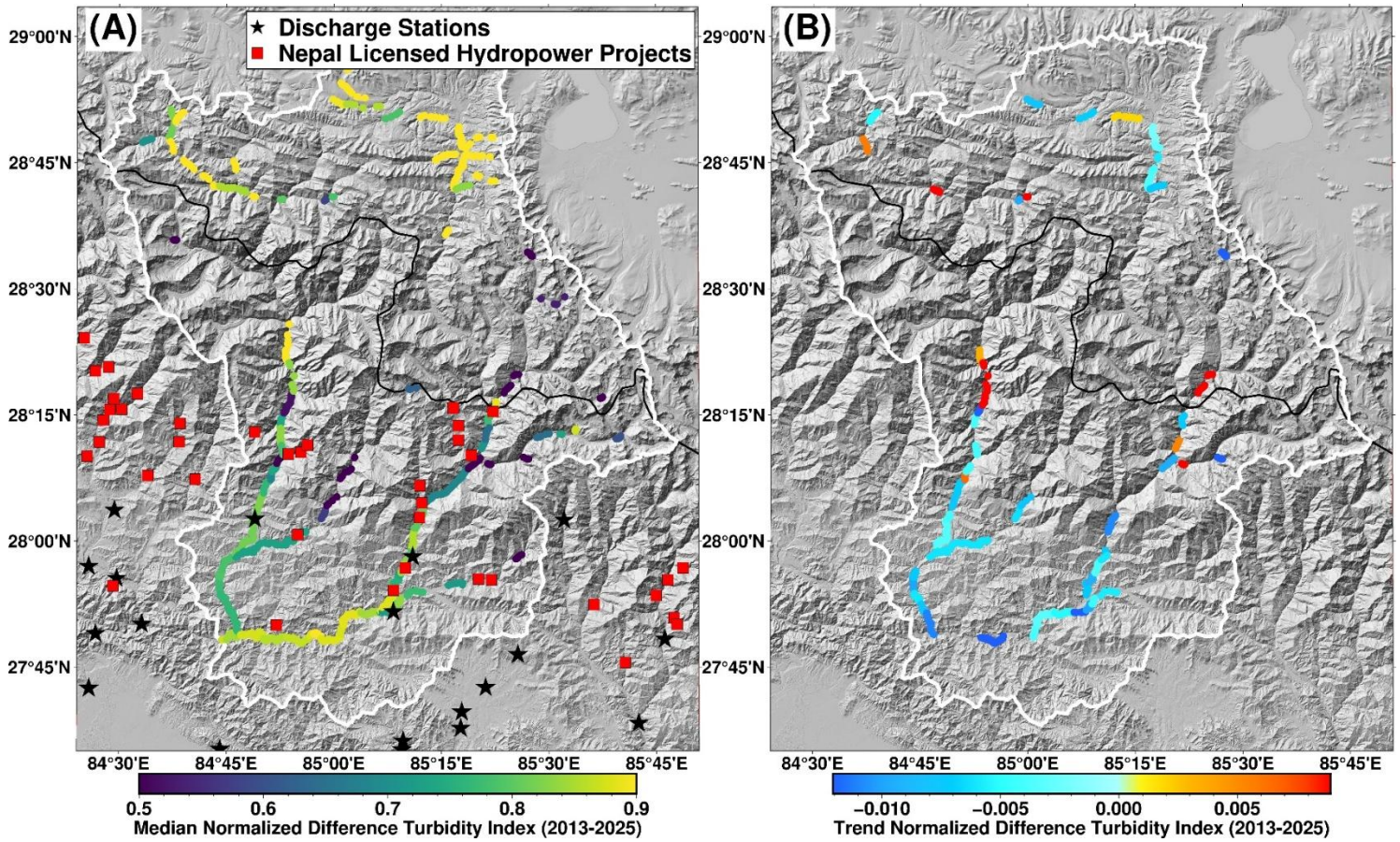
# Sediment Flux




**There have been large increases in sediment flux in many rivers**

https://www.science.org/doi/10.1126/science.abi9649

# Sediment Flux




**This change can also be seen at the local scale – water turbidity patterns are shifting**

Smith and Bookhagen, in prep

# DFG STRIVE Project

**The main goal of the ongoing work is understanding how climate change impacts alpine rivers**

**This includes changes in snowmelt, river temperature, water quality, and sediment flux**

# Workshop Goals

1. Learn about open-source hardware and how to build environmental sensors

2. Understand how to prepare data from in-situ sensors

3. Explain how remote sensing and satellite tools can help understand the environment

3. Gain hands-on experience with online data processing tools

5. Discuss ongoing research in High Mountain Asia related to water resources

# Workshop Materials

All materials and data can be found online:

https://github.com/tasmi/Workshop_Kathmandu_Feb2026

# Computer Setup

Please let us know if you had trouble with any set up!